

Discounting Disentangled

Moritz A. Drupp¹, Mark C. Freeman²,
Ben Groom³ and Frikk Nesje^{4*}

As the most important driver of long-term project evaluation, from climate change policy to infrastructure investments, the social discount rate (SDR) has been subject to heated debate among economists. To uncover the extent and sources of disagreement, we report the results of a survey of over 200 experts that disentangles the long-term SDR into its component parts: the pure rate of time preference, the wealth effect, and the real risk-free interest rate. The mean recommended SDR is 2.27 percent, with a range from 0 to 10 percent. Despite disagreement on point values, more than three-quarters of experts are comfortable with the median SDR of 2 percent, and over 90 percent find an SDR in the range of 1 to 3 percent acceptable. Our disentangled data reveal that only a minority of responses are consistent with the Ramsey Rule, the theoretical framework dominating discounting policy. Instead, experts recommend that governmental discounting guidance should be updated to deal with uncertainty, relative prices, and alternative ethical approaches.

Keywords: Social discount rate, project appraisal, expert opinions, disagreement.

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^{*1}Department of Economics, University of Kiel, Germany; mail: Drupp@economics.uni-kiel.de ² School of Business and Economics, Loughborough University, United Kingdom; mail: M.C.Freeman@lboro.ac.uk. ³ Department of Geography and Environment, London School of Economics, United Kingdom; mail: b.groom@lse.ac.uk. ⁴ Department of Economics and ESOP, University of Oslo, Norway; mail: frikk.nesje@econ.uio.no. We would like to express our deepest gratitude to our many survey respondents for their time and thoughts. We further thank Julius Andersson, Geir Asheim, Stefan Baumgärtner, Wolfgang Buchholz, Simon Dietz, Bård Harstad, Cameron Hepburn, Sverre Jensen, Antony Millner, Kian Mintz-Woo, Paolo Piacquadio, Martin Quaas, Thomas Sterner, Gernot Wagner and Marty Weitzman as well as seminar audiences at Bergen, Berlin, Camp Reinsehlen, EAERE 2015, Edinburgh, EEA 2015, EnvEcon 2015, Geneva, Gothenburg, Hamburg, Kiel, Oslo, Ulvön 2015 and ZEW for helpful discussions. We thank the LSE Research Seed Fund for financial support and Natalia Grinberg for research assistance. Drupp is grateful for financial support from the German National Academic Foundation, the DAAD, and the BMBF under grant 01LA1104C. Author for correspondence: Ben Groom.

‘social discount rate’ or ‘social discount factor’. We further performed an EconLit abstract-based search for the broader term ‘discount rate’ within the same journals. Correcting for scholars with multiple publications, and also discarding papers that did not pass a weak relevancy test (see Supplementary material A), our sample of potential experts includes 627 scholars.

There are some limitations to this selection strategy. First, by restricting the search to publications since the year 2000 to only capture scholars active in the current debate on social discounting, we potentially miss some relevant earlier contributors. Second, by selecting experts based on their publications, we necessarily include co-authors of relevant papers who are not themselves experts on discounting. Third, due to the rather generous weak relevancy test, we include a number of scholars who might not regard themselves as true experts on the issue. Fourth, we do not pick up relevant publications in the field that have used other terms to discuss discounting. Finally, we miss potentially relevant articles in lower-ranked journals. This may introduce a geographical bias into our sample by under-representing those from developing nations.

Despite these possible short-comings, the definition of expert that we deploy here is close to the one frequently used by policy makers, both in general policy contexts and in relation to social discounting.

Starting in May 2014, we sent out a link to the online survey (implemented in SurveyMonkey) via e-mail to all potential experts, and used three general rounds of reminders, each time slightly varying the subject line and motivation for answering the survey (Supplementary material B provides the initial e-mail text). In later rounds, we offered the option of completing the survey in a Word document or the e-mail itself to increase flexibility.

3. Survey Results

Table 1 provides summary statistics for each quantitative question in the survey as well as the imputed SRTP for each expert.

By November 2014 we had received quantitative and qualitative responses from 197 experts, including 12 who solely provided qualitative feedback containing important insights. We also received replies from 27 scholars explaining why they did not answer the survey and whose answer did not warrant inclusion as a qualitative

Table 1: Descriptive Statistics on Survey Results

Variable	Mean	StdDev	Median	Mode	Min	Max	N
Real growth rate per capita	1.70	0.91	1.60	2.00	-2.00	5.00	181
Rate of societal pure time preference	1.10	1.47	0.50	0.00	0.00	8.00	180
Elasticity of marginal utility	1.35	0.85	1.00	1.00	0.00	5.00	173
Real risk-free interest rate	2.38	1.32	2.00	2.00	0.00	6.00	176
Normative weight	61.53	28.56	70.00	50.00	0.00	100.00	182
Positive weight	38.47	28.56	30.00	50.00	0.00	100.00	182
Social discount rate (SDR)	2.27	1.62	2.00	2.00	0.00	10.00	181
SDR lower bound	1.12	1.37	1.00	0.00	-3.00	8.00	182
SDR upper bound	4.14	2.80	3.50	3.00	0.00	20.00	183
Social rate of time preference (SRTP)	3.48	3.52	3.00	4.00	-2.00	26.00	172
Number of quantitative responses							185
Number of qualitative responses							99
Number of responses used for analysis							197
Number of explained non-responses							27
Number of bias-check responses							38
Total number of responses							262

The SRTP is inputted from the individual determinants: the rate of pure time preference, and an interaction term of the growth rate of per-capita consumption and the elasticity of marginal utility of consumption.

response. The most common reason for such non-response was self-reported insufficient expertise, but it also included simply not having enough time or being unable to respond due to reasons of central bank confidentiality. Responses were also obtained after the survey closed from 38 previous non-respondents. This group is used to check for non-response bias.

Overall, we elicited 262 responses out of a pool of 627 potential experts. The response rate is 30 percent if we only consider the 185 quantitative responses. If we include all responses, the rate rises to 42 percent. Each is in line with comparable online surveys with economists (Necker 2014). Besides this, the sampling strategy was successful in obtaining responses from “blue ribbon” academic leaders on social discounting including 12 of the 13 experts of the Arrow et al. (2012) panel who advised the US EPA on this matter.

3.1. Quantitative Responses

3.1.1 Recommended Long-Term Social Discount Rate

In recent years, prominent experts such as Gollier (2012), Nordhaus (2008), Stern (2007) and Weitzman (2007) have proposed very different SDRs. Figure 1 (a) illustrates the extent of disagreement on the SDR for discounting real certainty-equivalent cash flows of a global public project with intergenerational consequences. The lowest recommendation is 0 and the highest 10 percent. However, the vast majority of experts provide point recommendations in the range of 0 to 4 percent, while the interval of 1 to 3 percent contains the point SDR recommendations of 68 percent of experts. The mean (median) value of the recommended SDR are 2.27 percent (2 percent), which are much lower than the corresponding values from Weitzman’s (2001) survey of economists of 3.96 percent (3 percent). Yet the most common single value recommended in these two different surveys is 2 percent. These results deviate significantly from the discount rates recommended in important recent guidelines, including the IPCC Fifth Assessment Report (IPCC 2014: 230).⁵

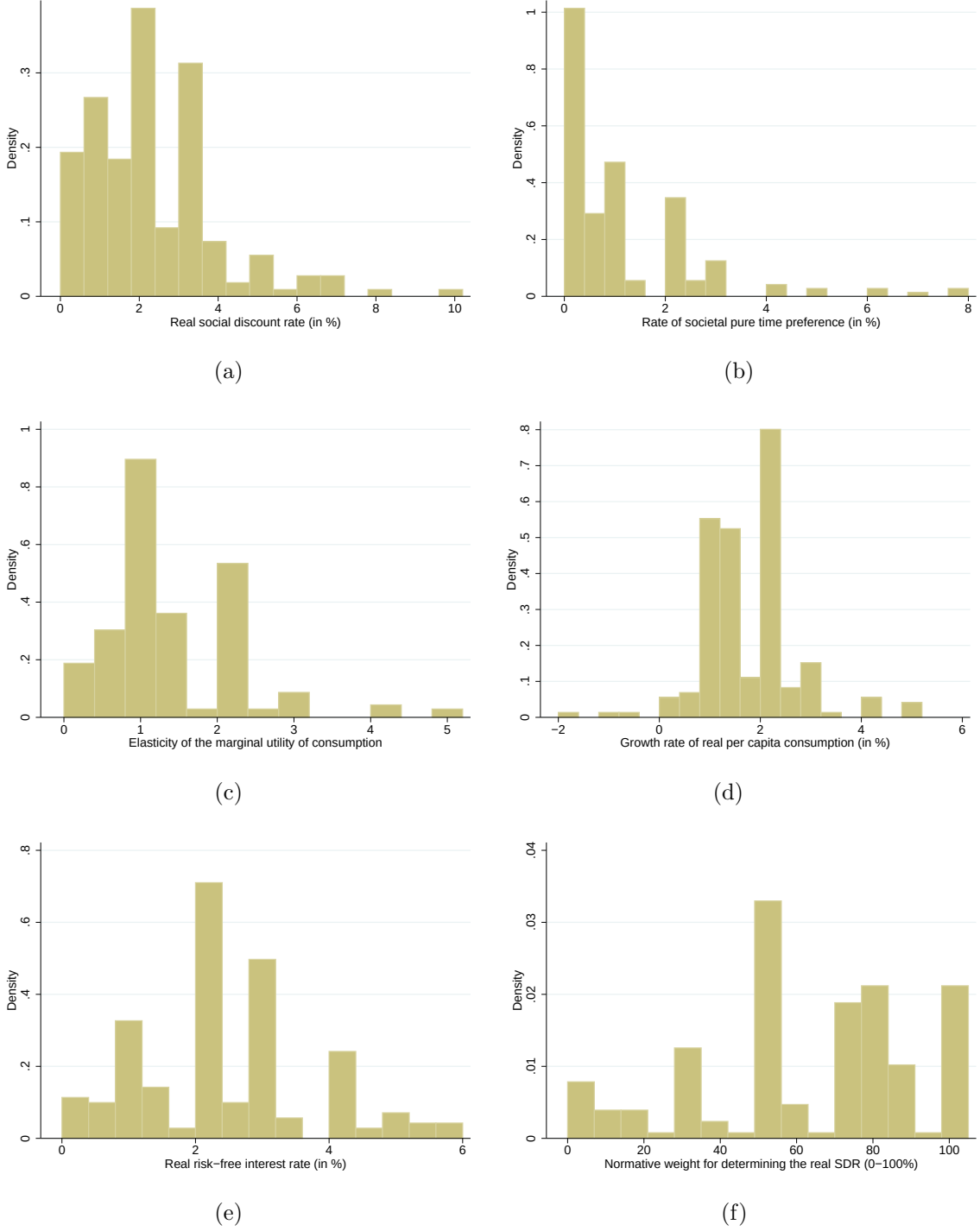


Figure 1: This figure provides histograms of expert recommendations and forecasts on determinants of discounting parameters. Figure (a) shows the real long-term SDR (in percent), (b) rate of societal pure time preference (in percent), (c) elasticity of marginal utility of consumption, (d) real growth rate of per capita consumption (in percent), (e) real risk-free interest rate (in percent), and (f) the normative weight for determining the SDR (in percent).

3.1.2 Rate of Societal Pure Time Preference

Positions on the rate of societal pure time preference, δ , have historically been the subject of intense disagreement. Luminaries of economics, such as Pigou, Ramsey and Harrod, believed that the well-being of each generation ought to be weighted equally, and so pure time preference should be zero. This view stems from their classical impartial Utilitarian philosophy. Disagreement surfaced again more recently with the publication of the Stern Review (Stern 2007), which took the same classical Utilitarian stance. Many alternative arguments exist for the use of a positive rate of societal pure time preference (e.g. Arrow 1999; Koopmans 1960; Nordhaus 2007). Figure 1 (b) shows substantial disagreement among experts on their chosen value for this parameter. As the modal value, a rate of zero is a focal point, and, if we include those responses that lie in the range of 0 to 0.1 percent, 38 percent take what might be called the Ramsey-Stern view. Yet, the distribution of responses is substantially right skewed with a median of 0.50 percent, a mean of 1.10 percent and a maximum recommendation of 8 percent. Based on these results, we cannot confirm the IPCC’s (2014: 229) conclusion that “*a broad consensus for a zero or near-zero pure rate of time preference*” exists among experts.

3.1.3 Elasticity of the Marginal Utility of Consumption

Settling on a value of the elasticity of the marginal utility of consumption, η , is an intricate affair. The reason is that it might capture vastly different concepts and thus lend itself to different interpretations. These are not only divided along the lines of normative (e.g., issues of distribution) and positive (e.g., preferences for consumption smoothing) determinants, but might also capture the inverse of the elasticity of intertemporal substitution or societal preferences for the aversion of consumption inequalities across space, time and also states of nature. All these rationales could have been used by different experts to inform their response, although the survey setting might reasonably have led respondents to primarily consider interpretations relating to an intertemporal context. Previous discussions in the literature point towards a range of 0.5 to 4 (Cowell and Gardiner 1999; Dasgupta 2008). The re-

⁵The results are closer to recent findings by Giglio et al. (2015) for long-term discount rates estimated from the housing market in Singapore and the UK, which imply risk-free discount rates of around 1 percent for 100-year claims on leaseholds.

sulting expert recommendations for elasticity of the marginal utility of consumption as presented in Figure 1 (c) are indeed widely dispersed, with a mean of 1.35 and a median and mode of 1. These values provide some support to the often made assumption of logarithmic utility.

3.1.4 Growth Rate of Real Per-Capita Consumption

Figure 1 (d) presents the results of our respondents' forecasts of the growth rate of real per-capita consumption, g . The overwhelming majority forecast a positive growth rate, with a mean of 1.7 percent and a median of 1.6 percent (cf. Table 1). This is close to the 2 percent growth rate of consumption per-capita in the western world for the last two centuries (Gollier 2012) and the 1.6 percent growth rate in GDP per-capita over the period 1900 to 2000 in non-OECD countries (Boltho and Toniolo 1999). Three experts project a negative growth rate, and 55 respondents forecast a lower growth rate than the IPCC's (2000) lower bound projection of 1.3 percent for the period from 1990 to 2100. 28 experts forecast a growth rate larger than 2 percent.

3.1.5 Real Risk-Free Interest Rate

Over the period 1900–2010, the global average real risk-free rate, r , was approximately 1 percent for bills and 2 percent for bonds (Dimson et al. 2011). For individual countries the rates are 1.1 percent for bills and 1.9 percent for bonds for the US, 0.8 percent and 2.0 percent for the UK and -0.5 percent and -0.6 percent for Japan (Dimson et al. 2011; see also Gollier 2012 for similar figures). The average response to our survey was 2.38 percent, with a standard deviation of 1.32 percentage points and a median value of 2 percent. The maximum forecast is 6 percent, while the minimum value, forecasted by three experts, is 0. The forecasted long-term global real risk-free interest rate according to our sample of experts is thus slightly higher than the estimated world average real risk-free rate of return on bonds over the past century.

3.1.6 Normative versus Positive Approaches

A central point of disagreement on the SDR concerns the question of whether normative issues, involving justice towards future generations, or positive issues, involving forecast average future returns to financial assets, or a mix of the two should determine the SDR (Arrow et al. 1996, 2014). Not everyone agrees that discounted utilitarianism is the correct ethical basis for intergenerational decision-making in the first place. For instance, some prefer deontological ethics which emphasize duties, while others prefer rights-based approaches. In line with this history of disagreement on the SDR, our intention for Question 5 is to establish the extent to which recommendations on the SDR are influenced by “*positive predictions and ethical judgments*” (Ricketts and Shoesmith 1992: 210–211). A clear finding from our data is that a large majority of experts (80 percent) think that both dimensions are relevant (see Figure 1 (f)). However, they generally recommend that governmental institutions should place greater weight on normative issues in determining the SDR; this has a mean (median) weighting of 61.53 percent (70 percent). When considering extreme weights, 14 percent (5 percent) of experts placed 0 (100 percent) weight on positive considerations, while 42 experts were divided equally between the two rationales; making this the modal response. These findings emphasize that setting the SDR requires both forecasts and value judgments.

3.2. Qualitative Responses

More than half of our respondents provided comments ranging from short remarks, such as “*risk matters*”, to explanations over multiple pages. The qualitative observations provide a rich body of evidence which shed light on various complexities of the theory and practice of social discounting. We group these individual comments into four main categories that address (i) individual survey questions Q1–Q5, (ii) technical issues, (iii) methodological issues, and (iv) concerns about limited expertise. Each category has multiple subcategories.

Table 2 provides an overview of the most common issues raised, including the number of experts that commented upon the respective issue and an example quote, sometimes edited for brevity. The five most often raised are: ‘declining discount rates and time-horizon’, ‘uncertainty’, ‘substitutability and environmental scarcity’, ‘heterogeneity and aggregation’, and ‘comparison to the Ramsey Rule’.

Table 2: Overview of Qualitative Responses

Issue	N	Exemplary quote
Q1: Growth rate	14	I foresee a very bright economic future with a continued 2 percent growth rate for the coming century.
Q2: Pure time preference	10	I see no reason to treat generations not equally.
Q3: Elasticity of marginal utility	12	The elasticity of marginal utility of consumption is heterogeneous, and using a single value is a crude simplification.
Q4: Real risk-free interest rate	8	There is no interest rate for 100 year horizon (to my knowledge).
Q5: Normative vs. positive	16	The components of the SDR are overwhelmingly normative in nature.
Declining discount rates and time horizon	20	I am more comfortable with declining discount rates [...] due both to declining time preference rates and to uncertainty about future consumption growth.
Heterogeneity and aggregation	19	Ideally, the input for our [social welfare function] would be a utility function that allows for heterogeneous preferences.
Opportunity cost of funds	8	SDRs should reflect the social opportunity cost of borrowed funds.
Project risk	6	We would have to consider very carefully the risk structure of the investment to get a correct discount rate.
Relative prices of non-marketed goods	20	If future costs/benefits accrue e.g. to environmental amenities, I would argue for a very low discount rate, based on an expectation of increasing relative prices for these goods.
Uncertainty	20	We need to admit that the current state of the world is full of uncertainties. [Yet] most uncertainties are neglected, and sometimes few remain when these are considered most important, [...] or easiest to accommodate.
Alternatives to discounting	15	Instead of imposing a [social welfare function] and calculate the corresponding optimum, it is ‘better’ to depict a set of feasible paths of consumption, production, temperature, income distribution, etc. and let the policy maker make a choice.
Comments on the survey	14	The search for THE discount rate, if that is your project, is deeply flawed.
Confidence intervals	8	I would also insist on providing confidence intervals.
Ramsey Rule	17	My discount rate is less than implied by the Ramsey rule because I use the extended rule, incorporating uncertainty.
Role of experts	7	I really think economists have very little special expertise in knowing the ‘right’ number. These parameters should be chosen in an open, iterative way with an eye toward understanding the consequences of different choices.
Limited confidence	13	Please ignore my response to Q4: I don’t have the knowledge to make a meaningful forecast.
Limited expertise	5	I am not a real expert on these issues.

3.3. Non-Response Bias and Representativeness

We followed several standard procedures to test for the existence of non-response bias. First, from December 2014 to April 2015 we contacted via e-mail and telephone 60 randomly selected non-respondents. This allowed us to obtain a further 38 responses, with 14 of these experts providing qualitative data and 24 giving reasons for their initial non-responses.⁶

Second, our sample includes quantitative responses from 11 of the 13 “blue ribbon” experts on social discounting from the Arrow et al. (2012) panel that advised the US EPA.⁷

Third, we divide the sample between early and late responses (Dalecki et al. 1993, Necker 2014), defined by the subsample of 58 experts that directly responded to the first e-mail and those that answered a reminder e-mail. Potential non-response bias is checked for by comparison of the mean and median responses of each of these groups, which are reported in Table 3. While there are differences regarding some of the discounting determinants, we find that there are no statistically significant differences in the mean and median SDR values between different groups.

A further common measure for potential non-response bias is to consider groups by gender and location (Necker 2014).⁸ We find that male experts selected into responding to our survey relative to the non-response group (91 percent versus 81 percent). The proportions of respondents and non-respondents are balanced in terms of characteristics such as being a full Professor (49 percent versus 48 percent) and average year of Ph.D. completion (1993.6 versus 1993.7). Experts currently based in Europe selected into responding (49 percent of respondents versus 32 percent of non-respondent), which provides some indicative evidence for a potential geographical

⁶Reasons include having insufficient time (11 times) or expertise (10 times). The high proportion of respondents stating to be no expert suggests self-selection of true experts into responding to the initial survey.

⁷A twelfth panel member initially provided qualitative evidence only, but stated after the survey was completed that he would “follow the view of the median panelist.”

⁸Personal data are obtained, where possible, from the experts’ own web pages. We collect information on current continental location, gender, whether or not they have a professorial title, and year of Ph.D. graduation as a proxy for (academic) age. We are able to identify 89 respondents from Europe, 80 from the Americas and 16 from the Rest of the World (RoW), who work exclusively from Asia and Oceania. We have 167 male respondents, while only 16 women gave quantitative answers to our questionnaire. Approximately half our sample are professors and the mean year of Ph.D. graduation is 1994.

Table 3: Comparison with Non-Respondents and Arrow et al. (2012) Experts

	g	δ	η	r	Normative	SDR	SDRmin	SDRmax
<i>Results from the 185 quantitative responses</i>								
Mean	1.70	1.10	1.35	2.38	61.53	2.27	1.12	4.14
Median	1.60	0.50	1.00	2.00	70.00	2.00	1.00	3.50
N	181	180	173	176	182	181	182	183
<i>Results from the 14 randomly selected previous non-respondent responses</i>								
Mean	1.63	1.46	1.23	1.96	71.36	2.02	1.01	3.09
Median	1.50	1.00	1.00	1.75	75.00	2.00	0.63	3.00
N	12	12	8	12	12	13	14	13
<i>Results from 11 of the 13 Arrow et al. (2012) panel experts</i>								
Mean	1.80	0.60	1.51	2.66	57.27	2.62	1.30	4.00
Median	2.00	0.50	1.50	3.00	50.00	3.00	1.00	4.00
N	11	11	11	11	11	11	11	11
<i>Results from the 58 early responses</i>								
Mean	1.49	0.73	1.47	2.26	60.14	1.99	0.92	3.68
Median	1.50	0.38	1.50	2.00	50.00	2.00	0.75	3.00
N	58	58	56	58	58	58	58	58
<i>Results from the 127 late responses</i>								
Mean	1.80	1.27	1.29	2.44	62.18	2.40	1.21	4.35
Median	1.80	0.90	1.00	2.00	70.00	2.00	1.00	4.00
N	123	122	117	118	124	123	124	125

bias. This may have led to a slight underestimation of the mean SDR given the propensity of Europeans to be more normative than non-Europeans, as discussed in Section 4.1.1.

We present a more detailed analysis of non-response bias checks in Supplementary material C. Overall, our findings do not suggest substantial and systematic unidirectional non-response biases for SDR recommendations.

6. Conclusion

We have presented evidence from a survey of over 200 experts on the determinants of the long-term real social discount rate (SDR). This is, perhaps, the single most important driver of any cost-benefit analysis evaluating long-term public projects. We find that the mean (median) recommended SDR of our experts is 2.27 percent (2 percent). While there is considerable disagreement between respondents on point recommendations, which range from 0 to 10 percent, more than three-quarters of those surveyed would find a SDR of 2 percent acceptable. More than 90 percent are comfortable with a SDR somewhere in the interval of 1 percent to 3 percent.

A key innovation of our survey is that we not only elicit responses on the appropriate SDR itself, but also on individual discounting determinants: recommendations on the rate of pure time preference and the elasticity of marginal utility of consumption, as well as predictions of long-term per-capita consumption growth and the average real risk-free rate of interest.

Based on this disentangled data, we show that the simple Ramsey Rule, which is widely accepted as a *“useful conceptual framework for examining intergenerational discounting issues”* (Arrow et al. 2012: 3) and can be found in governmental guidelines on cost-benefit analysis across the world, cannot explain the responses of the majority of our experts. The conclusion that more complex models for social discounting are required is supported by the rich body of qualitative responses we received. Many of our respondents provided comments relating to a number of extension and alternatives to the simple Ramsey Rule approach. These include issues such as uncertainty, heterogeneity, relative prices of non-marketed goods as well as other societal evaluation approaches that policy guidance on social discounting should consider to ensure efficient and equitable decisions on long-term public projects.

Further inspection of the data on individual discounting determinants yields findings that transcend their value as potential inputs to governmental discounting guidelines. First, we find that the modal value of the pure rate of time preference is zero, but with a mean (median) of 1.1 percent (0.5 percent), our results cannot confirm the IPCC’s (2014: 229) conclusion that *“a broad consensus for a zero or near-zero pure rate of time preference”* exists among experts. Second, our data suggest that the IPCC should consider lower growth scenarios in future assessments. Third

the median and mode recommended elasticity of marginal utility of consumption of 1 provides some support for the often made assumption of logarithmic utility. Lastly, we find that there exists considerable disagreement between experts on the relative importance of normative and positive approaches to discounting. Most report that the SDR should reflect both, highlighting that these previously accepted categories overly polarize more nuanced expert views. Engaging with both disagreement about values and uncertainty over forecasts is therefore an essential task for informing decision-making on long-term public projects.

Overall, our findings lead us to the conclusion that the prominence of the Ramsey Rule needs to be revisited and that current policy guidance concerning social discounting and the evaluation of long-term public projects needs to be updated. While not uncontentious, our survey points to a long-term global SDR for certainty-equivalent cash flows of 2 percent.